

KAVINRAJA D

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PROFESSIONAL SUMMARY

Results-driven AI/ML engineering student with hands-on experience in designing, developing, and deploying machine learning models and algorithms. Proficient in implementing ML solutions using modern frameworks and cloud platforms, with demonstrated ability to evaluate model performance and maintain comprehensive documentation. Skilled in building data pipelines and collaborating with cross-functional teams to integrate models into production environments.

EDUCATION

B.Tech, Artificial Intelligence and Machine Learning, Saveetha Engineering College, Chennai

Aug 2022 – May 2026

CGPA: 8.5/10

Relevant Courses: Data Structures, Machine Learning, Deep Learning, DBMS, Algorithm Implementation

SKILLS

Programming Languages: Python (pandas, NumPy, scikit-learn), SQL, Java

ML Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, Keras

Cloud & MLOps: AWS SageMaker, Streamlit Cloud

Infrastructure & DevOps: Basic CI/CD principles, Infrastructure-as-code concepts

Model Evaluation: Accuracy, Precision, Recall, Performance Analysis

Data Processing: Data pipeline development, Feature engineering, Data augmentation

Generative AI & LLM: Prompt Engineering, Gemini 1.5 Flash, Language model API integration

Development Tools: Git, VS Code, Jupyter, Documentation practices

Core Competencies: Analytical thinking, Problem-solving, Technical communication, Client-focused collaboration, Quality assurance

EXPERIENCE

Gen AI - Python & Automation Intern, Sasken Technologies Ltd.

Jun 2025 – Present

- Designed and developed **end-to-end ML models** using **TensorFlow** and **PyTorch** for automated deep learning solutions, contributing to client challenge resolution.
- Implemented and tested **ML algorithms** with focus on model performance evaluation including accuracy and precision metrics.
- Integrated **language model APIs** into internal production tools, enhancing user interaction capabilities and system efficiency.
- Collaborated with engineering teams to understand mobile AI application architecture and deployment lifecycle from backend to production.

PROJECTS

MouthMap – Computer Vision ML Pipeline

Dec 2024 – Apr 2025

- Developed and deployed a production-ready **CNN + BiLSTM model** for lip movement recognition, achieving high accuracy through systematic algorithm implementation and testing.
- Built comprehensive **data preprocessing pipelines** using OpenCV to process and align 1000+ video samples with transcript data, ensuring data quality and model reliability.
- Created and maintained thorough **documentation** of model parameters, preprocessing steps, and deployment workflows for production integration.
- Deployed scalable solution using **Streamlit Cloud**, implementing user-friendly interface for video processing and live model testing.

Document Intelligence OCR System

Mar 2025 – Apr 2025

- Implemented **CRNN-based OCR algorithms** for automated extraction of register numbers and subject codes from scanned documents, focusing on accuracy and efficiency.
- Applied **data augmentation techniques** and feature engineering to improve model performance on handwritten text recognition tasks.
- Developed batch processing capabilities using **Streamlit** framework, enabling efficient handling of multiple document files with comprehensive testing workflows.
- Documented model evaluation results and preprocessing methodologies following best practices for production deployment.

COMPETITIONS & TECHNICAL ACHIEVEMENTS

- SIH 2024 (ISRO)** — Implemented deep learning algorithms for SAR image processing using advanced CNN architectures, demonstrating proficiency in computer vision model development.
- Amazon ML Challenge 2024** — Enhanced OCR pipeline accuracy by 10% through systematic feature engineering, model optimization, and performance evaluation techniques.
- IITM Shastra Hackathon (Winner)** — Led cross-functional ML development team in implementing supervised learning algorithms (SVMs, decision trees), achieving 88% model accuracy through statistical analysis and systematic testing.

CERTIFICATIONS

AWS Certified AI Practitioner

Deep Learning with TensorFlow

Oracle Certified AI Foundations